16 April 2026 Research Assignment 1
Foundations of Data Analytics & Data Science.

Section A:

1. Database is an organized data of information it is stored in a structured way. To be easily accessibile, managed & updated.
We use it in businesses & organizations to help store large amounts of information securely.
Real-world examples will be:
- Online Shopping
- Banking systems
- Social Media

2.
- Data warehouse is a central place where we store multiple database This is used mainly for analysis & reporting, it is designed to help Organizations in understanding trends, patterns & strategic decisions.

- It differes from regular a database cause it is used for analysis rather than daily operations.
e.g It supports reporting & decision making
- It is Historical, aggregated data overtime
- Combines data from multiple systems.

Companises use data warehouse because they provide a clear, big-picture view of the business.
- It brings data together from different departments e.g finance, marketing, sales.

3.
Data Lake is a large storage system that holds huge amounts of raw data in its original format until it is needed.

- This data can be structured : Tabel format
~~ot~~ or unstructured : Videos, images, emails.

- How it differs from data warehouse :
Data warehouse stores structured, cleaned & original data ~~whilto~~ whilst
Data Lake stores raw data in any format.

When ~~why~~ would a business prefer a data lake :
- When it needs to handle large volumes of different data & doesn't yet know how that data will be used.

4.
Data Mart is smaller & it is focused on specific department, team or business function. It only includes relevant data in a particular area. Makes it easier, faster for users to access what they need.

- How it relates to a data warehouse is it is usually derived from a data warehouse.
e.g Data warehouse : Stores comprehensive, enterprise-wide data
Data Mart : extracts & organizes a portion of data for a specific purpose.
- It is typically used by :
Sales team analyzing revenue & customer trends, Marketing teams tracking campaign performance, finance teams reviewing budgets & forecasts.

- Why they are useful :
Improve performance by reducing the amount of data being queried.
Simplify access to relevant information.
Enable faster decision-making within departments.

5.

Structured Data : Highly organized & stored in a fixed format e.g Rows, columns like a table. " customer database with fields like name, email, phone number & account balance.

Semi-Structured Data: It Does not follow a strict table format but it contains organization through e.g tags, labels or keys.

Unstructured Data : It has no predfined format or organization. It is the most complex type & often requires advanced tools & process & analyze. e.g Images, videos, social media.

6. Data Modelling is; the process of creating a structured representation of data & how it relates within a system

- It is important: Because it ensures that a system handles data efficiently & accurately
- Two types of Data Models:

1. Conceptual data model : It it a high-level model that shows the main entities & relationships in a system without going into technical details.
2. Logical Data Model :

This model provides more detail by defining attributes, keys & relationships

7. Data mining is the process of analyzing large sets of data to discover hidden patterns, trends, relationships, or useful insights.

- How is is used: In business to better understand customers, improve operations, & make smarter decisions. E.g helps companies identify customer behavior & preferences.

Real world examples:

- Retail & Online Shopping: Companies use data mining to analyze customer purchase history & browsing behavior.

DMS (Database Management System)

8. Data Management is software that allows users to create, store, manage & retrieve ~~data~~ data in a database.

- It helps to maintain data intergrity, control access and handle large volume of data efficiently.

Popular DBMS platforms & their use cases:

1. MySQL:

- It is an open-source relational DBMS that stores data in tables with rows & columns.

Use cases: Ideal for websites & web applications, such as e-commerce platforms.

2. PostgreSQL:

- It is an advanced open-source relational DBMS that supports complex queries & large-scale applications.

Use Case: Suitable for applications requiring complex data processing, such as financial systems, analytics platforms or geographical information system.

3. Oracle Database:

- It is a powerful, enterprise-level DBMS known for high performance, scalability and strong security features.

Uses case: Commonly used by large organizations for mission-critical systems like banking, enterprise resource planning (ERP), & large-scale transaction processing

4. MongoDB:

- It is a NoSQL DBMS that stores data in flexible, & It is designed for handling large volumes of unstructured or semi-structured data.

Use case: Ideal for modern applications such as mobile apps, real-time analytics & content platforms where data structure may change frequently.

9. ~~Et~~ ETL (Extract, Transform, Load) is a process used in data analytics to collect data from different sources, to clean & organise it. Store it in a centralized system for analysis.

1. Extract: This step involves gathering data from various sources such as spreadsheets, ~~the~~ API's or external system. E.g pulling sales data from retail system.

2. Transform: This step the extracted data is cleaned, formatted, & converted into a consistent structure. This may include removing duplicates, fixing errors or missing values. E.g Converting all ~~data~~ dates into the same format & merging customer records from different systems.

3. Load: The final step is loading the transformed data into a target system e.g database or data warehouse, where it can be accessed for reporting & analysis. E.g storing the cleaned data in a central warehouse for business intelligence tools to use.

This is important in data analytics because raw data is often messy, inconsistent, & spread across multiple systems. It ensures that data is e.g accurate & reliable, consistent across sources, ready for analysis & reporting, centralized for easy access.

10. Data storage formats are used depending on how data needs to be stored, processed & shared.

1. CSV (comma-seperated values): A simple text format where data is ~~Description~~ stored in rows & columns, seperated by commas. Use it for simple datasets, data exchange between systems, or when working with spreadsheets & basic analysis.

2. JSON (JavaScript Object Notation): It is a semi-structured format that stores data as key-value pairs. When to use for Web applications, APIs, or when working with semi-structured or nested data.

3. Parquet: A columnar, binary storage format optimized for big data processing. When to use it - for larger-scale analytics, especially in big data environments like lakes or frameworks.

3. Reinforcement Learning → It involves training a model through trail & error, where it learns by receiving rewards or penalties based on its actions. Eg Robotics & Self-driving cars.

13. Deep Learning is a specialized area of Machine Learning (ML) that uses large, complex models called neural networks to learn patterns from data.

- How it relates to ML & AI :

AI → The broad field of creating machines that can perform tasks requiring human intelligence.

ML → A subset of AI where systems learn from data

Deep Learning → A subset of ML that uses multi-layered neural networks to learn from large amounts of data.

Neural Networks : Are computational models inspired by the human brain.

How they work is by :

1. Input data is fed into the network, e.g an image or text.
2. Data passes through layers, Each neuron processes the data using weights (importance values) and applies a mathematical function.
3. Learning happens through adjustment, the network compares its output to the correct answer & adjusts the weights to improve accuracy (this process is called training).
4. Output is produced → after training, the network can make predictions on new data.

14. Natural Language Processing (NLP) is a field within Artificial Intelligence that enables computers to understand, interpret, & generate human language (both written & spoken). It allows machines to interact with people in a natural way.

Real world Applications of NLP :

1. Voice assistants, tools like Siri, Google Assistant use NLP to understand spoken commands & respond appropriately.

2. Chatbots & Messaging: Customer service chatbots on websites & apps use NLP to understand user questions & provide answers.

3. Autocorrect & predictive text, when typing on your phone, features like autocorrect and next-word prediction rely on NLP to understand what you're writing & suggest corrections or completions.

15. ~~A~~ Artifical intelligence model designed to understand, generate, & work with human language. It is trained on massive amounts of text data (books, websites, ~~artifi~~ articles. & learns patterns in lanuage so it can perform tasks like answering questions, writing contents, translating languages & summarizing informations. LLMs are typically built using deep learning techiques, especially neural networks called transformers.

How LLMs differ from standard Machine Learning Models.

A standard Machine Learning (ML) model is usually trained for a sepecific task

- Predicting house prices
- Classifying emails as spam or not
- Recognizing images.

These models:

- Use smaller, task-specific datasets.
- Require retraining or redesign for new tasks
- Focus on structured or narrowly defined data

In contrast, LLMs:

- Are trained on very large & diverse ~~database~~ datasets.
- Can perform multiple tasks without retraining (called generalization)
- Work primarily with unstructured text data.
- Can generate human-like lanuage.

Two well-known LLMs & their uses.

1. GPT-4
- Developed by OpenAI
- Used in chatbots (like ChatGPT), content writing, coding assistance, and tutoring
- Can understand complex prompts & generate detailed, human-like responses.
2. BERT (Bidirectional Encoder Representations from Transformers)
- Developed by Google
- Mainly used for understanding language rather than generating it.
Powers search engines, helping improve search results & understand user queries more accurately.

1b. Generative AI is a type of artifical intelligence that can create new content rather than just analyze or predict. This include text, images, music, code, or even videos.

How Generative AI works in 3 steps:
1. Training on large datasets; Model is trained on massive amounts of data (texts, images) & learn patterns, structures, & relationships within that data.
2. Learning patterns using neural networks; It uses deep learning models (often transformer-based for text or diffusion models for images) to understand how data is structured.
3. Generating new content; When given a prompt (like a question or instruction) ~~that~~ the model predicts & generates new content based on what is learned

3 Generative AI tools & what they do.
1. ChatGPT
- Developed by OpenAI
- Generates human-like text
- Used for answering questions, writing emails, coding, tutoring.
2. DALL-E
- Also developed by OpenAI

- ~~Also~~ Creates images from text descriptions
- Useful for design, marketing visuals, & creative ~~ab~~ artwork.

3. Midjourney
- An independent AI tool
- Generates highly detailed & artistic images from prompts.
- Popular among designers & digital artists.

17. A training dataset is the collection of data used to teach an AI model how to perform a task. e.g (like text, images or numbers) & often the correct outputs.

Why the quality of training data matters:
- The quality of training data directly determines how well the AI model performs.

High-Quality data should be:
- Accurate (correct & reliable)
- Relevant (related to the task)
- Diverse (covers different scenarios & cases)
- Balanced (not skewed toward one outcome)

What can go wrong with biased or incorrect data: Could lead to
1. Bias & unfair outcomes, if certain groups are underrepresented or misrepresented, the model may produce e.g unfair hiring recommendations
2. Inaccurate predictions: Incorrect or noisy data can cause the model to learn wrong patterns, leading to poor performance
3. Overfitting or underfitting: Poor-quality or limited data can make the model memorize instead of learn (overfitting).
4. Loss of trust: Users may lose confidence in the system if it produces unreliable or unfair outputs

18. ~~to~~ Computer vision is a field of Artificial Intelligence that enables machines to interpret & understand visual information from the world e.g images & videos.

How machines see & interpret images: Series of steps

1. Image capture & representation, An image is broken down into a grid of pixels pixels contain numerical values, color & intensity.
2. Feature extraction: Using models like neural networks, the system detects patterns such as edges, shapes, textures & color.
3. Pattern recognition: The model compares these features to what it learned during training to identify objects, faces or scenes.
4. Interpretation & decision-making: The system assigns meaning (e.g this is a car or this is a person") and acts on it.

Two real-world use cases

1. Facial recognition: Used in smartphones & security systems.
   e.g unlocking your phone using your face.
2. Self-driving Cars: Autonomous vehicles use Computer Vision to detect roads, pedestrians, traffic signs & other vehicles.

Section C:

19. API (Application Programming Interface): Is a set of rules that allows different software systems to communicate & share data with each other.

Simple real-world analogy

Think of API like a resturant waiter:

- You (the user) choose what you want from the menu.
- The waiter (API) takes your request to the kitchen.
- The kitchen (system/server) prepares your food.
- The waiter brings the finished dish back to you.

Why APIs are important: Is Are essential in modern software cause they make systems more connected, flexible & scalable

Enable intergration : Different applications can work together. Eg A mobile app can use a payment API to process ~~trant~~ transactions.
Save development time : Developers don't need to build everything from scratch
Support scalability : Systems can grow & add new features without rebuilding everything.
Power modern apps & services : Many everyday apps rely on APIs to functions as : Social Media logins, Weather updates, Online payments.

20. API key is a unique code (a string of letters, & numbers) used to identify & authenticate an application when it makes requests to an API.
Purpose of An API key :
- Authenticate requests, Confirm that the request is coming from a valid user or application
- Control access : Limit what data or features the requester can use.
- Track usage : Monitor how often the API is used (for billing, rate limits or analytics).
Simple analogy : API key like a Membership card at a gym :
- You show your card to enter, The gym verifies you're allowed in, They can track how often you visit.
How to store & protect API keys should be handled carfully because they grant access to services.
• Keep them secret : Never share them publicly or expose them in code repositories.
• Use environments variables : Store keys outside your main code eg in environment configs.
• Use secure storage : Such as encrypted key vaults or secret mangers
• ~~Dont~~ Restrict permissions : Give the key only the access it absaulutely needs
• Rotate keys regularly : Change them ~~pe~~ periodically to reduce risk.